

The Clock-Host Candidate in One Page: What Stands, What Died, What Decides

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The theory. A single-metric khronometric clock $n_\mu = -\partial_\mu \tau / |\partial \tau|$ hosting one MOND scalar φ , matter minimally coupled:

$$\mathcal{L} = \sqrt{-g} \left[R - K_B (\nabla_\mu n_\nu \nabla^\mu n^\nu - \nabla_\mu n_\nu \nabla^\nu n^\mu) - c_2 (\nabla \cdot n)^2 + (c_{14} - K_B) J_\mu J^\mu + 2(2 - K_B) J^\mu \partial_\mu \varphi - (2 - K_B) \mathcal{J}(Y) - \xi^2 (D^2 \varphi)^2 \right] + \mathcal{L}_m,$$

$J^\mu = n^\nu \nabla_\nu n^\mu$, $Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \varphi \partial_\nu \varphi$, $\mathcal{J} \rightarrow \frac{2}{3} Y^{3/2} / \tilde{a}_0$ in deep MOND, D^2 the leaf Laplacian, ξ a screening length ($\xi \gtrsim 0.8$ pc from the ephemerides). The acceleration scale is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ *fitted*: the action has a zero mode that leaves κ free (kappa_closure k01–k03); measured 0.551 ± 0.043 (distance-free) and 0.465 ± 0.076 (BTFR), degenerate at 2σ with Milgrom’s 0.461 and Verlinde’s 0.482; a 1.2% measurement separates them.

What stands, each by a committed computation with a Lean certificate of its algebra (real_research/clock_2026, L279–L289; kappa_slot_2026 SW04–SW07):

- **Lensing = dynamics is derived, not assumed** (L279): the traceless equation carries no scalar term, so $\Psi = \Phi$ including the MOND part; slip $3.5\text{--}3.9 \times 10^{-7}$ at 30 kpc for $10^{11} M_\odot$; $G_N = G/(1 - c_{14}/2)$ reproduced.
- **PPN settled** (L280): $\alpha_1 = -4c_{14}$, $\alpha_2 = c_{14}[c_{14}(1 + 2c_2) - c_2]/[c_2(2 - c_{14})]$; passes at $c_{14} \leq 8 \times 10^{-7}$ or on the equal-speed locus $c_2 = c_{14}/(1 - 2c_{14})$, $c_{14} \in [1, 2.5] \times 10^{-5}$; the corner the record had used failed α_2 by $12\times$. Cassini, $c_T = c$, GW170817 hold on the record.
- **Healthy on every background**: uniform ellipticity for all field strengths (L281); exactly two propagating scalar degrees of freedom, khronon and MOND scalar, with lapse, shift and Φ constrained, in vacuum, rolling and dust-loaded (L287, CK06).
- **The scalar is cold where it must be** (L282): $c_s^2 = (2 - K_B)(J_Y - \beta_0)/[|K_2| + \text{clock mixing}]$, $\beta_0 = (2 - K_B)/(2 - c_{14})$ — the same coefficient that fixes the deep-MOND normalisation makes the cosmological scalar exactly pressureless; galactic $c_s \approx 430$ km/s.
- **Cosmological background from the action** (L283): $G_{\text{cosmo}} = G/(1 + 3c_2/2)$ so BBN needs $c_2 < 0.074$; scalar law $a^3 F_Q = \text{const.}$
- **Direction-blind EFE programme closed** (SW04–SW07): the smoothed-total-switch action dies on α_1 ; the candidate predicts Gaia wide binaries $\gamma_v = 1.000$ against AQUAL’s 1.16–1.23.

What died this week, by the same machinery. The candidate’s dark sector was the clock–scalar field’s own energy from the roll of $Q = n \partial \varphi$ (the AeST mechanism; no particles). *Quadratic well* $K_2(Q - Q_0)^2$: its stiff a^{-6} companion forces $|K_2|Q_0^2 \geq 1.8 \times 10^{23} H_0^2$ at BBN, and at that floor the roll mode grows at $10^4\text{--}10^6 H_0$ on 10 kpc with no screening length that heals it (L285, L287). *Exponential wall* at $Q_0 = H_0$: a long-wavelength instability of the dust-loaded clock sector, hundreds per e-fold ($10^2\text{--}10^8 H_0$) until $a_* = 0.5/0.2/10^{-3}$ at $k = 0.01/0.1/1$ Mpc $^{-1}$, confirmed by the exact Minkowski dispersion to 0.1% (L288). *The killer is the clock coupling*: a dust coupled through the metric alone is linearly healthy at the same densities (L289) — and is ordinary CDM, which KiDS forbids in galaxies ($\leq 14\%$ halo).

What any dark carrier must do (L289, isothermal retention in deep-MOND wells against the record’s constraints): $c_s(z=0) \in [200, 800]$ km/s to leave galaxies pure MOND yet fill clusters, while the forest needs $c_s(z=3) \leq 9.5$ km/s: a homogeneous carrier must heat by $\geq 445\times$ in c_s^2 since $z = 3$ ($c_s^2 \propto a^p$, $p \geq 4.4$), or carry the MOND scalar’s field-dependent pressure $c_s \propto (g_N/a_0)^{1/4}$, whose only known inertia is the clock-coupled term that died. No viable carrier is in hand; the CMB third peak and the cluster factor of two are open.

What decides. (i) Gaia DR4 wide binaries, preregistered (Amendment 11): Arm B $\gamma_v = 1.000$ flat, ceilings 1.0450/1.0300, killed from above only; Arm A (AQUAL) 1.16–1.23. (ii) A deep-MOND lensed rotator at $z \approx 2$: the BTFR zero point to ± 0.13 dex separates the flat law (0.00) from the Λ CDM-emergent scale (+0.33); the DESI $w(z)$ mapped through the framework’s pressure law gives +0.01 to +0.06 dex at $z = 2.5$ (L273–L275). (iii) κ to 1.2%.

Corrections kept on the record this week: L283’s physical reading withdrawn by L284, L284’s withdrawn by L285, L285’s screening claim withdrawn by L287, L286’s diagnosis retracted by L287 (an operator bug). Four of nine lanes were corrected by the next lane; every one is committed with its FAIL lines. *Every number above is re-read from a committed script.*